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Competing in a supply chain via full-refund and no-refund customer returns policies

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ABSTRACT

In this paper, we address the problem in which a manufacturer in a supply chain supplies a product to two competing retailers. Both retailers, facing customer returns, must decide on prices for the product. The manufacturer is a Stackelberg leader in the supply chain and the two retailers decide prices after the manufacturer announces the wholesale price. One retailer is well established, offering a full-refund policy, while the other retailer is a new entrant into the market place. The new entrant retailer needs to decide whether to implement a full-refund policy or a no-refund policy, in order to compete with the well-established retailer under two games: Nash game and Stackelberg game. We show that customer returns policies can be used as competing marketing strategies. Numerical examples are included to illustrate the major results of the paper. The analysis generates insights about how the degree to which the customer accepts the no-refund product affects the new entrant retailer's choice of customer returns policies in a competitive market.

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1. Introduction

Handling customer returns has become critical as the volume of products flowing back from customers increases (Guide et al., 2006). Using survey data, the National Retail Federation reports the value of products being returned as around an estimated $194 billion, or 8.12% of total sales, in 2010. Toktay (2003) states that customer return rates range from 5% to 9% of sales for most retailers. Mostard and Teunter (2006) report that for catalog retailers, return rates on fashion items "are generally around 35–40%, but can be as high as 75%" of sales. When product returns occur, handling the returned merchandise can impose substantial costs on retailers. This has a substantive impact on retailer profitability, because returned products are used products to some extent and are expected to have lower value than new products. Chen and Bell (2009) show that a retailer who fails to account for customer returns in price and ordering decisions will suffer excessive inventory and a significant drop in profitability. In fact, Lawton (2008) points out that it is estimated that the U.S. electronics industry spent $13.8 billion to repackage, restock, and resell returned products in 2007. Among returned products, only about 5% of returns were because a product was truly defective. Instead, most consumers reject products for other reasons, such as the device being too confusing to use. Grimaldi (2008) notes that it is estimated that managing product returns costs U.S. companies over $200 billion annually.

Usually, a retailer receives returned products from customers through a specified customer returns policy. In practice, there is no law that requires any retail merchant to have a specific refund, credit or exchange policy. Customer returns policies can range from "100% money back guarantees" to "no refunds or exchanges" whatsoever. Several studies have examined how customer returns policies affect the customer purchase and return decision, and the consequences for retailers' profits (for example, Che, 1996; Davis et al., 1995, 1998). In practice, many retail stores in North America implement a liberal customer returns policy, because refusing to accept the returned product is viewed as risking offending a customer and losing a stream of revenue from repeated sales (Davis et al., 1995). The Hudson's Bay Company, Sears, and Wal-Mart, for example, offer full refund policies for most of the products they sell. For customers, of course, a full-refund customer returns policy can eliminate the risk of having to keep an unwanted product.

Some retailers, however, do not allow customer to return products. Many retailers conduct periodic "sales" with an "all sales are final" statement (essentially, the no-refund case), implying that the price mark-down is accompanied by a loss of product return ability. It is common that Wal-Mart and Hudson's Bay do not provide refunds for food or fashion product effectively disallowing the return of these products. When retailers provide no-refund policies, they can save costs of having to keep returned product and costs that are associated with handling customer returns.

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The customers, on the other hand, will take the risk of keeping mismatched product.

As compared to products with a full-refund policy, customers perceive the same products sold with a no-refund policy as being of lower value, due to the risk of having to keep the unwanted products (Heiman et al., 2005). In practice, some retailers offer a full-refund policy to customers on products sold at a higher price in regular stores, while offering no-returns policy on final sales at a lower price in "factory outlets." Davis et al. (1995) provide a rationale for higher prices for returnable products: "money back guarantees may allow the retailer to charge higher prices because the reduction in the risk from the products being a poor match with their tastes may increase a consumer's willingness to pay." In this paper, we model that customers perceived values are low for the product with a no-refund policy and are high for the product with a full-refund policy.

In this paper, we examine how a new entrant retailer can use a full-refund returns policy or a no-refund returns policy to compete with an established retailer who offers a full-refund customer returns policy. The existing literature suggests that in a duopoly setting money-back guarantees (MBGs) can be utilized by firms with a low likelihood of returns to signal product quality to uninformed customers (for example, Moorthy and Srinivasan, 1995; McWilliams, 2012). McWilliams (2012) shows that MBGs can help the low-quality retailer in a duopoly model. In addition, some products, such as durable products, have a relatively stable returns rate, due to the nature of the products, suggesting that their returns have little relation to the brand of the retailer. These previous studies, however, do not include the conditions under which a retailer can use customer returns policies as a means of competition with other retailers. In our paper, we will identify these conditions.

2. Literature review

In the literature, returns are typically assumed to be a proportion of product sold (sales), which implies that if more items are sold, more products will be returned from customers (for example, Vlachos and Dekker, 2003). To mitigate the negative impact of customer returns on the retailer's profit, some research has factored customer returns into pricing and ordering decisions. Based on the assumptions that a fixed percentage of sold products will be returned and that products can be resold at most once in a single period, Vlachos and Dekker (2003) investigate optimization of order quantities for a newsvendor-style problem in which the retail price is exogenous. Chen and Bell (2009) examine the impact of customer returns on a firm's pricing and ordering decisions as well as on the firm's profit. Xiao et al. (2010) consider a manufacturer and a retailer supply chain where the retailer faces customer returns. They focus on the coordination mechanism in a supply chain considering product returns. Su (2009) studies the impact of returns polices (full-refund and partial-refund) on supply chain performance and proposes a mechanism to coordinate the supply chain in the presence of customer returns. Shulman et al. (2009) describe how consumer purchase and return decisions are affected by a seller's pricing and restocking fee policy. They identify conditions under which it is (or is not) optimal to provide product fit information to consumers. None of these papers considers how a retailer can use different returns policies as a competitive marketing strategy.

Although accepting customer returns reduces the retailer's profit, it provides a valuable service to customers. Not accepting customer returns saves the retailer expenditure, but customers take the risk of keeping unwanted product after the purchase. Offering different customer returns policies allows retailers to differentiate themselves and discriminate prices, such that the retailer can charge a higher price for a product that allows for return but a lower price for the same product without the possibility of return. Customers also have different preferences on customer returns policies. Some customers, who are price sensitive, prefer a low price at the expense of a no-refund policy, while other customers prefer a liberal return policy, even if the product price is higher. Heiman et al. (2005) show empirically that a product with a more lenient customer returns policy results in a higher value, as perceived by customers. Chen and Bell (2012) consider the options of a retailer selling product through a returnable channel, a non-returnable channel, or a dual channel consisting of a returnable channel and a nonreturnable channel. They show that using different returns policies to segment a market and differentiate prices is a good way to enhance the firm's profit, while serving distinct customer bases with various needs and purchase preferences. Making use of the results from the empirical study (Heiman et al., 2005) and as in Chen and Bell (2012), in this paper, we suggest that customers perceive a higher value for the product with a full-refund policy and a lower value for the product with a no-refund policy. We investigate how a new entrant retailer can implement a full-refund customer returns policy or a no-refund customer returns policy to compete with an established retailer offering a full-refund customer returns policy, in a supply chain setting.

In many industries, retailers distribute substitutable products and compete in markets that are limited in extent. The game-theoretic models are typically based on the assumption that demand depends on price. The nature of this dependence, however, varies from model to model (for example, Savaskan and Van Wassenhove, 2006). Anderson et al. (1996) provide a review of the analytical literature on product and price competition between firms.

Economists have been interested in differentiated markets since the late 1920s. Hotelling (1929) uses a linear city model, while Salop (1979) uses a city circular model (Spatial model) to formulate the problem in which a firm competes directly only with its neighbor. They assume that conditional on neighbor prices, fluctuations in prices of more distant competitors have no effect on own sales. Consumer preferences determine the location of the ideal product. Spatial models are now widely used in marketing and economics research literature (for example, Bakos, 1991; Pinkse et al., 2002; Chun and Kim 2005; Moorthy and Zhang, 2007). Hauser and Shugan (1983) examine how an existing brand should respond to a competitive new product entry. In the present paper, we use Hotelling's linear city model to model two competing retailers facing customer returns at locations a unit distance apart. The model of two competing retailers locating a unit distance apart has been widely used in the marketing and economics literature. Both retailers are supplied by a single manufacturer. One is an established retailer offering a full-refund policy, while the other is a new entrant. We examine under what conditions a full-refund policy or a no-refund policy is a good option for the new entrant retailer, and whether Nash game or Stackelberg game is a better equilibrium between the two competing retailers, from the perspectives of the new entrant retailer and as well as those of the manufacturer; we also consider supply chain efficiency.

For the duopoly competing model in the presence of customer returns, Shulman et al. (2011) examine the pricing and restocking fee decisions of two competing firms facing customer returns and selling horizontally differentiated products to customers who have heterogeneous tastes for the products. They find that in the partial refund returns policies, restocking fees are more severe when consumers are less informed about product fit and when consumers place a greater importance on how well products' attributes fit with their preferences. McWilliams (2012) has developed a model for high- and low-quality retailers. He finds that the low-quality retailer gains, while the high-quality retailer loses, relative to cases in which MBGs are not offered, when consumer's reservation price is heterogeneous.

Here, we will examine the conditions under which a new entrant into the market can use a full-refund or a no-refund returns policy to compete with an existing retailer with a full-refund policy, if both retailers sell the same products. We model product value as higher with a full-refund policy and lower with a no-refund policy. Our two competing firms differentiate by location. We focus on the role of returns policies as a marketing competitive advantage.

Therefore, the paper contributes to the literature by examining the impact of customer returns, returns policies, and customer purchasing behavior on pricing decisions in a supply chain that consists of two competing retailers and a manufacturer. The paper also generates new insights into the consideration of customer returns policies for a new entrant in a competitive market.

The rest of the paper is organized as follows. In the following section, the model framework and assumptions are described. Sections 4 and 5 discuss the case in which the new entrant competes with a full-refund policy and with a no-refund policy, respectively. Section 6 identifies the boundary conditions for when the entrant retailer should use a full-refund policy or use a no-refund policy as a competing marketing strategy. Numerical examples are given in Section 7. Conclusions and the direction of future work are given in Section 8. All proofs are in the Appendix.

3. Model framework and assumptions

In this paper, we consider a supply chain in which a manufacturer supplies a product to two competing retailers. Both retailers, facing customer returns, must set prices for the product. The manufacturer is a Stackelberg leader in the supply chain and the two retailers decide prices after the manufacturer announces the wholesale price. One retailer is established, and offers a full-refund returns policy, while the other retailer is a new entrant. The new entrant retailer needs to decide which customer returns policy, full-refund or no-refund, should be implemented as a marketing strategy to compete with the established retailer. In practice, retailers offer various customer returns policies, including full-refund, partial refund, or no-refund, to balance customer satisfaction and the loss resulting from customer returns. On the other hand, customers have different purchasing behaviors based on familiarity with the product, sensitivity to the product's price, and probability of returning the product (Chen and Bell, 2012). In this paper, we use the framework of Hotelling (1929), because of its analytical tractability, considering two retailers who are located at a certain distance from each other, which is normalized to one unit. They may have different marketing power. We examine how a new entrant retailer can compete with the established retailer by using different customer returns policies.

Let two retailers be at the two ends of a unit length, as shown in Fig. 1. We assume that the customer's transportation cost is t $(0<t<1)$ per unit distance. A customer whose position is x to the retailer 1 incurs a transportation cost of xt if he purchases the product at retailer 1 and incurs a transportation cost of $(1-x)t$ if he purchases the product at retailer 2.

The manufacturer, as a Stackberg leader, should determine the wholesale price (w) in anticipation of the retailers' selling prices, given a production cost of c per unit. After the manufacturer announces w, retailers decide selling prices. Without loss of generality, we assume that in a local market, retailer 1 is an established retailer offering a full-refund customer returns policy to its customer. Retailer 2 is a new entrant, needing to decide which customer returns policy, a full-refund or a no-refund policy, should be implemented to compete with retailer 1. In reality, there are two different possible situations for the new entrant. For example, a well-known brand retailer (for example, Sears) will open a new store to compete with an existing retailer (for example, Wal-Mart) in a local market. In such a case, the retailer's size is competitive to that of the existing retailer. The new entrant can compete with the existing retailer in a Nash game. In the literature, several papers model the new entrant to compete with the existing competitors in a Nash game (for example, Xie and Sirbu, 1995; Shiau and Michalek, 2009; Choi et al., 1990). On the other hand, if the new entrant is a retailer with such as a small size in business, it can be a pricing follower in the market to compete with the existing retailer. Such case is likely a common case in business (for example, Steiner, 2010; Ginevičius and Krivka, 2008). Several papers discuss both games for the comparison purpose (for example, Xie and Sirbu, 1995; Choi et al., 1990). Since both games are possible in practice, in this paper, we thus consider two equilibrium games – Nash equilibrium and Stackelberg equilibrium – between the two retailers in deciding their retail prices. In a Nash equilibrium (denoted with a superscript N), both retailers have equal marketing powers, representing the case that two retailers are competitive. In a Stackelberg equilibrium (denoted with a superscript S), retailer 1 is the Stackelberg leader, who has an established business or has a large size in business, while retailer 2 is a follower who might have a small size in business.

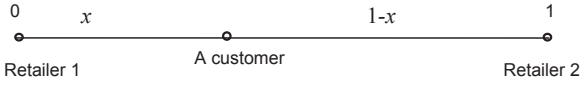


Fig. 1. Positions of two competing retailers.

Following the studies of Heiman et al. (2005) and Chen and Bell (2012), we assume that the reservation price of the customer for the product with a full-refund policy is v, following the uniform distribution. Without affecting the results, for simplicity, as the widely used assumption in the literature, we assume that v is in the range of [0, 1] (for example, Chiang et al., 2003). The reservation price of the customer for the product with a no-refund policy is θ, where $0<\theta<1$ (Chen and Bell, 2012). θ reflects the discount that customers expect to compensate for the lack of returns service, or the degree to which the customer accepts the no-refund product on an aggregate level. When the retailers accept customer returns, although the returned product has a salvage value, the retailers incur a cost per unit in handling customer returns. This handling cost includes the cost of the staff time for receiving a returned product and updating records. We define the net salvage value, s_r per unit as the revenue from salvaging of the product minus any cost of handling the returned product. s_r can be interpreted as the efficiency of handling customer returns. We follow the assumption in McWilliams (2012) in which he assumes the identical salvage values and customer returns handling costs (thus the identical net salvage values) at two retailers who sell a product with different qualities. Without affecting the results of the paper while keeping analytical tractability, we assume that both retailers have the same efficiency in handling in customer returns (s_r) due to selling the identical product. In the numerical example section, we will conduct a sensitivity study to illustrate the impact of the different net salvage value on pricing decisions and profits of the two retailers. To ensure that the manufacturer benefits from selling products to retailers instead of from salvaging them, it is reasonable to assume that $1>c>s_r$. Also, we assume that $s_r>0$ so that retailers have the motivation to handle customer returns, which gives $1>c>s_r>0$.

Anderson et al. (2009) and Hess and Mayhew (1997) conducted empirical studies and showed that the number of returns has a strong positive linear relationship with the quantity sold, while Vlachos and Dekker (2003) assumed that customer returns were a fixed proportion of quantity sold. Following the results of these empirical studies and the assumption used in the literature, we assume that the retailers face customer returns that are proportional to their sales, i.e., the customer returns rate is α. α reflects the portion of product that does not fit or match the expectation of customers, which mainly depends on the nature or the quality of

Table 1
Notations.

Notation	Description
p_{r1} and D_{r1}	Retailer 1's price and demand, respectively
p_{r2} and D_{r2}	Retailer 2's price and demand when it offers a full-refund policy
p_{n2} and D_{n2}	Retailer 2's price and demand when it offers a no-refund policy
Π_{Ri} and Π_M	Retailer i's profit, $i=1$, 2 and manufacturer's profit, respectively
t	Transportation cost
w	Manufacturer's wholesale price per unit
c	Manufacturer's production cost per unit
α	Customer returns rate
s_r	Net salvage value

the product (for example, Davis et al., 1995; Hsiao and Chen, 2012). We thus assume that both retailers expect same α, because they sell the same product. The customer returns rate aggregately reflects the level of the product that cannot meet customer's needs. We also assume that returned products cannot be resold in this single-period problem due to the need for inspection and repackaging. Not reselling returned product is also in the line with the product manufacturer's preference, because the quality of returned product is uncertain, which could result in high customer dissatisfaction. We denote price, demand, cost, and profit of the retailer with a full-refund policy with the subscript r and of the retailer with a no-refund policy with the subscript n. We also use subscripts R and M to denote the retailers and the manufacturer, respectively. We summarize notations that will be used in this paper in Table 1.

We first discuss the case in which both retailers offer a full-refund policy.

4. A full-refund policy at both retailers

Let p_{1r} and p_{2r} be the prices set by retailers 1 and 2. A customer who is x units away from retailer 1 will buy a product from retailer 1 if he has a positive utility, i.e., $1-p_{r1}-tx>0$, which gives $x<(1-p_{r1})/t$. He will buy the product from retailer 2 if $1-p_{r2}-t(1-x)>0$ or $1-x<(1-p_{r2})/t$. He prefers to purchase from retailer 1 rather than from retailer 2 if $1-p_{r1}-tx>1-p_{r2}-t(1-x)$ or $x<(t+p_{r2}-p_{r1})/(2t)$. If $(1-p_{r1})/t<(t+p_{r2}-p_{r1})/(2t)$ or $t>2-p_{r1}-p_{r2}$, the market is partially covered, suggesting that only when transportation cost is sufficiently low will the market be fully covered. Let this threshold value of transportation cost be t_1, i.e., $t_1=2-p_{r1}-p_{r2}$. Thus, the demands for retailers 1 and 2 are

When $t>t_1$,

$$\begin{cases} D_{r1}=(1-p_{r1})/t, \text{ and} \\ D_{r2}=(1-p_{r2})/t. \end{cases} \tag{1}$$

when $t\le t_1$,

$$\begin{cases} D_{r1}=(t+p_{r2}-p_{r1})/(2t), \text{ and} \\ D_{r2}=(t+p_{r1}-p_{r2})/(2t). \end{cases} \tag{2}$$

Retailers maximize their profits (Π_{R1},Π_{R2}) by setting retail prices (p_{r1},p_{r2}):

$$\Pi_{R1}=D_{r1}[p_{r1}(1-\alpha)-w]+\alpha s_r D_{r1}, \tag{3}$$

and

$$\Pi_{R2}=D_{r2}[p_{r2}(1-\alpha)-w]+\alpha s_r D_{r2} \tag{4}$$

The manufacturer decides the wholesale price in anticipation of retailers' pricing decisions. Its profit function is

$$\Pi_M=(w-c)(D_{r1}+D_{r2}) \tag{5}$$

we now discuss two cases, a partially covered market (high transportation cost, where $t>t_1$) and a fully covered market (low transportation cost, where $t\le t_1$).

4.1. Partially-covered market

Eqs. (1), (3) and (4) suggest that when both retailers offer a full-refund customer returns policy and the transportation cost t is sufficiently high ($t>t_1$), one retailer's choice of selling price will not affect the other. We denote the wholesale price for this case as w_{rr}. Thus, we have the following result:

Proposition 1. *If both retailers offer full-refund policies and if $t>t_1$, there exists an equilibrium, and the optimal pricing decisions for the supply chain (w^*_{rr},p^*_{r1},p^*_{r2}) are given by*

$$\begin{cases} w^*_{rr}=(1-\alpha+\alpha s_r+c)/2, \text{ and} \\ p^*_{r1}=p^*_{r2}=\frac{3(1-\alpha)-\alpha s_r+c}{4(1-\alpha)}. \end{cases} \tag{6}$$

With (6), we have $D_{r1}=D_{r2}=[1-\alpha+\alpha s_r-c]/[4(1-\alpha)t]$. In the presence of customer returns, only when $\alpha<\overline{\alpha}_{rr}=(1-c)/(1-s_r)$, both retailers can generate demands from the market and demands decrease with transportation cost (t).

With (6), the profits of the retailers and the manufacturer are

$$\begin{cases} \Pi_{R1}(p^*_{r1},p^*_{r2},w^*_{rr})=\Pi_{R2}(p^*_{r1},p^*_{r2},w^*_{rr})=\frac{(1-\alpha+\alpha s_r-c)^2}{16(1-\alpha)t} \text{ and} \\ \Pi_M(p^*_{r1},p^*_{r2},w^*_{rr})=\frac{(1-\alpha+\alpha s_r-c)^2}{4(1-\alpha)t}. \end{cases} \tag{7}$$

With p^*_{r1} and p^*_{r2} in (6), t_1 is given by

$$t_1=\frac{1-\alpha+\alpha s_r-c}{2(1-\alpha)} \tag{8}$$

Eq. (8) gives when $\alpha<\overline{\alpha}_{rr}$, $t_1\ge 0$. We also have $\partial p^*_{r1}/\partial\alpha=\partial p^*_{r2}/\partial\alpha>0$, $\partial D_{r1}/\partial\alpha=\partial D_{r2}/\partial\alpha<0$, $\partial w^*_{rr}/\partial\alpha<0$, $\partial\Pi_M(p^*_{r1},p^*_{r2},w^*_{rr})/\partial\alpha<0$, and $\partial\Pi_{R1}(p^*_{r1},p^*_{r2},w^*_{rr})/\partial\alpha=\partial\Pi_{R2}(p^*_{r1},p^*_{r2},w^*_{rr})/\partial\alpha<0$. As customer returns rate (α) increases, if both retailers offer a full-refund customer returns policy and $t>t_1\ge 0$, both retailers raise p^*_{r1} and p^*_{r2} and the manufacturer reduces w^*_{rr}. Demands for both retailers decrease, however, which results in a decrease in the profits of both the manufacturer and the retailers. Further, $\partial t_1/\partial\alpha<0$, suggesting that a larger α, a smaller transportation cost (t) can lead to the case of partially-covered market. When $\alpha>\overline{\alpha}_{rr}$, the market cannot be fully covered.

4.2. Fully-covered market

When $t\le t_1$, where t_1 is given in (8), the market is fully covered. Demand function in (2) suggests that when $t\le t_1$, the total demand of the two retailers is equal to 1. In such a case, the manufacturer's profit (Π_M) increases with its wholesale price (w).

A positive t_1 requires $\alpha<\overline{\alpha}_{rr}$. We consider two equilibrium games, Nash equilibrium and Stackelberg equilibrium, between the two retailers. In Nash equilibrium, each retailer considers the other's pricing decision in pricing decisions and announces retail prices (p^N_{r1},p^N_{r2}) simultaneously. In Stackelberg equilibrium, after retailer 1 announces the selling price (p^S_{r1}), retailer 2 decides the selling price (p^S_{r2}). With demands given in (2) and profits functions given in (3)–(4), for a given w, we have the results as follows.

Corollary 1. *If both retailers offer a full-refund policy and if $t \le t_1$, the market is fully covered.*

(i) *If the two retailers compete in Nash game, there exists a unique optimal retail price pair $(p_{r1}^{N*}, p_{r2}^{N*})$, which is given by*

$$p_{r1}^{N*} = p_{r2}^{N*} = \frac{t(1-\alpha) - \alpha s_r + w}{1-\alpha}, \tag{9}$$

(ii) *if the two retailers compete in Stackelbeg game, there exists a unique optimal retail price pair $(p_{r1}^{S*}, p_{r2}^{S*})$, which is given by*

$$\begin{cases} p_{r1}^{S*} = \frac{3t(1-\alpha) - 2\alpha s_r + 2w}{2(1-\alpha)} \text{ and} \\ p_{r2}^{S*} = \frac{5t(1-\alpha) - 4\alpha s_r + 4w}{4(1-\alpha)}. \end{cases} \tag{10}$$

with demands in (2) and retailers' profits in (3)–(4), (9) and (10) give

In Nash game,

$$D_{r1}^{N*} = D_{r2}^{N*} = 1/2 \tag{11}$$

and

$$\Pi_{R1}^{N*} = \Pi_{R2}^{N*} = \frac{t(1-\alpha)}{2} \tag{12}$$

In Stackelberg game

$$\begin{cases} D_{r1}^{S*} = 3/8, \text{ and} \\ D_{r2}^{S*} = 5/8; \end{cases} \tag{13}$$

$$\begin{cases} \Pi_{R1}^{S*} = \frac{9t(1-\alpha)}{16} \text{and} \\ \Pi_{R2}^{S*} = \frac{25t(1-\alpha)}{32}. \end{cases} \tag{14}$$

The results suggest that for both Nash and Stackelberg games between the two retailers, demands and profits for the two retailers are independent of the wholesale price, since the wholesale price has the same impact on p_{r1}^{N*} and p_{r2}^{N*} [from (9)], p_{r1}^{S*} and p_{r2}^{S*} [from (10)]. If retailer 2 is a new entrant to the market, as a Stackelberg follower, it can set a lower price and gain a higher market share and profit, as compared to those of retailer 1. Comparing two games between two retailers, we can conclude

Corollary 2.

$D_{r1}^{S*} < D_{r1}^{N*}$, $D_{r2}^{S*} > D_{r2}^{N*}$, $\Pi_{R1}^{S*} > \Pi_{R1}^{N*}$, *and* $\Pi_{R2}^{S*} > \Pi_{R2}^{N*}$

Corollary 2 suggests that if retailer 2 offers a full-refund policy to compete with retailer 1, retailer 1 prefers to be a leader, while retailer 2 prefers to be a follower. Considering the manufacturer's wholesale price (w_{rr}), (9), and (10), we have

Proposition 2. *When both retailers offer a full-refund policy and when $t \le t_1$,*

(i) *if the two retailers compete in Nash game, the best action for the manufacturer is to set w_{rr}^{N*} at*

$$w_{rr}^{N*} = [(2-3t)(1-\alpha) + 2\alpha s_r]/2, \tag{15}$$

the optimal pricing decisions for the two retailers $(p_{r1}^{N*}, p_{r2}^{N*})$ become

$$p_{r1}^{N*} = p_{r2}^{N*} = (2-t)/2, \tag{16}$$

(ii) *if the two retailers compete in Stackelbeg game, the manufacturer will set w_{rr}^{S*} at*

$$w_{rr}^{S*} = [(8-15t)(1-\alpha) + 8\alpha s_r]/8; \tag{17}$$

the optimal pricing decisions for two retailers $(p_{r1}^{S*}, p_{r2}^{S*})$ become

$$\begin{cases} p_{r1}^{S*} = (8-3t)/8 \text{ and} \\ p_{r2}^{S*} = (8-5t)/8. \end{cases} \tag{18}$$

With (15) and (17), the manufacturer's respective profits in the two games are

$$\Pi_M^{N*} = [(2-3t)(1-\alpha) + 2\alpha s_r - 2c]/2 \tag{19}$$

and

$$\Pi_M^{S*} = [(8-15t)(1-\alpha) + 8\alpha s_r - 8c]/8 \tag{20}$$

In addition, for both Nash and Stackelberg games between the two retailers, we see that retailers' pricing decisions and corresponding demands are independent of customer returns rate (α) and the net salvage value (s_r). This implies that when retailer 2 knows that retailer 1 implements a full-refund policy and the transportation cost is low ($t \le t_1$), if it decides to compete with a full-refund policy, the pricing decisions for both retailers depend only on transportation cost (t). Therefore, both retailers can decide the retail price without considering α and s_r.

A positive profit at the manufacturer requires $w_{rr}^{N*} > c$ and $w_{rr}^{S*} > c$, i.e., $t < \overline{t_1^N} = 2[(1-\alpha) + \alpha s_r - c]/[3(1-\alpha)]$ and $t < \overline{t_1^S} = 8\ [(1-\alpha) + \alpha s_r - c]/\ [15(1-\alpha)]$, respectively. Since $\overline{t_1^N} > t_1$ and $\overline{t_1^S} > t_1$, we can conclude that $\partial w_{rr}^{N*}/\partial\alpha < 0$, $\partial w_{rr}^{S*}/\partial\alpha < 0$, $\partial \Pi_M^{N*}/\partial\alpha < 0$, and $\partial \Pi_M^{S*}/\partial\alpha < 0$, suggesting that as customer returns rate (α) increases, in both equilibriums, the manufacturer should reduce the wholesale price and incur a loss in profit. Further, the profits of the two retailers also decrease with α.

Comparing two games between two retailers, we can conclude

Corollary 3. *As compared to Nash game, in Stackelberg game, the manufacturer sets lower wholesale prices and gains less profit. Also, the supply chain efficiency is lower.*

Corollaries 2 and 3 imply that if retailer 2 is a new entrant as a follower and uses a full-refund policy to compete with retailer 1, both retailers are better off, but the manufacturer incurs a significant loss in profit. The above analysis raises a cautionary note for the manufacturer. Since the manufacturer cannot influence the profits of the two retailers by setting a lower wholesale price, it needs other incentives (such as sharing a portion of its profit) to incentivize retailer 2 to compete with retailer 1 in Nash equilibrium, rather than in Stackelberg equilibrium as a follower.

We now discuss the case if retailer 2 competes with retailer 1 via a no-refund policy.

5. A full refund policy at retailer 1 and a no-refund policy at retailer 2

When retailer 1 offers a full-refund policy while retailer 2 does not allow for customer returns, a customer who is x units away from retailer 1 will buy a product from retailer 1 if $x < (1-p_{r1})/t$. He will buy the product from retailer 2 if $\theta - p_{n2} - t(1-x) > 0$ or $1-x < (\theta - p_{n2})/t$. He prefers to purchase from retailer 1 rather than from retailer 2 if $x < (1-p_{r1}-\theta+t+p_{n2})/(2t)$. If $(1-p_{r1})/(t) < (1-p_{r1}-\theta+t+p_{n2})/(2t)$ or $t > t_2 = 1-p_{r1}+\theta-p_{n2}$, the market is partially covered. Only when the transportation cost (t) is sufficiently low ($t \le t_2$) can the market be fully covered. The demand for retailers 1 and 2 are

When $t > t_2$,

$$\begin{cases} D_{r1} = (1-p_{r1})/t, \text{ and} \\ D_{n2} = (\theta - p_{n2})/t. \end{cases} \tag{21}$$

When $t \le t_2$,

$$\begin{cases} D_{r1} = (1-p_{r1}-\theta+t+p_{n2})/(2t), \text{ and} \\ D_{n2} = (t-1+p_{r1}+\theta-p_{n2})/(2t). \end{cases} \tag{22}$$

Each retailer maximizes its profit by setting its own retail price:

$$\begin{cases} \Pi_{R1} = D_{r1}[p_{r1}(1-\alpha)-w] + \alpha s_r D_{r1}, \text{ and} \\ \Pi_{R2} = D_{n2}(p_{n2}-w). \end{cases} \tag{23}$$

The manufacturer maximizes profit by deciding the wholesale price:

$$\Pi_M = (w-c)(D_{r1} + D_{n2}) \tag{24}$$

5.1. Partially-covered market

The retailers' decisions on selling prices will not affect each other if $t > t_2$. We denote the optimal wholesale price as w_{rn}^*. Thus, we have the following result:

Proposition 3. *When the transportation cost (t) is sufficiently high ($t > t_2$), if retailer 1 offers a full-refund policy and retailer 2 offers a no-refund policy, there exists Nash equilibrium and the optimal pricing decisions for the supply chain ($w_{rn}^*, p_{r1}^*, p_{n2}^*$):*

$$\begin{cases} w_{rn}^* = \frac{(1-\alpha)(1+\theta)+\alpha s_r+c(2-\alpha)}{2(2-\alpha)}, \\ p_{r1}^* = \frac{(1-\alpha)(2+\theta)+[(1-\alpha)-\alpha s_r](3-2\alpha)+c(2-\alpha)}{4(2-\alpha)(1-\alpha)}, \text{ and} \\ p_{n2}^* = \frac{(1-\alpha)+\theta(5-3\alpha)+\alpha s_r+c(2-\alpha)}{4(2-\alpha)}. \end{cases} \tag{25}$$

With demand in (21), we have

$$D_{r1} = \frac{2(1-\alpha)^2 + (1-\alpha)(1-\theta) + \alpha s_r(3-2\alpha) - c(2-\alpha)}{4(2-\alpha)(1-\alpha)t}$$

$$D_{n2} = [(3-\alpha)\theta - (1-\alpha) - \alpha s_r(3-2\alpha) - c(2-\alpha)]/[4(2-\alpha)t]$$

In addition,

$$t_2 = \frac{(1-\alpha)(1+\theta) + \alpha s_r - c(2-\alpha)}{4(1-\alpha)} \tag{26}$$

Since $D_{r1} > D_{n2}$, only when

$$\alpha < \overline{\alpha}_1 = \frac{5-\theta-3s_r-c-\sqrt{(1-\theta)^2 + (c-s_r)[6-9s_r+2\theta+c)}}{4(1-s_r)}, \tag{27}$$

both retailers can generate demand from the market and demand decreases with the transportation cost (t). Comparing to $\overline{\alpha}_{rr}$ in Section 4.1, we find that $\overline{\alpha}_{rr} > \overline{\alpha}_1$, suggesting that it allows a larger returns rate (α) for having market shares for both retailers when both retailers offer a full-refund policy. A positive t_2 requires $\alpha < \overline{\alpha}_{rn} = (1+\theta-2c)/(1+\theta-c-s_r)$. Also, $t_1 > t_2$ if $0 < \alpha < \overline{\alpha_2}$ while $t_1 < t_2$ if $\overline{\alpha_2} < \alpha < \min(\overline{\alpha}_{rr}, \overline{\alpha}_{rn})$, where $\overline{\alpha_2} = (1-\theta)/(1-\theta+c-s_r)$. The profits of retailers and the manufacturer are

$$\begin{cases} \Pi_{R1}(p_{r1}^*, p_{n2}^*, w_{rn}^*) = \frac{[(1-\alpha)(3-2\alpha-\theta)+\alpha(3-2\alpha)s_r-c(2-\alpha)]^2}{16(1-\alpha)(2-\alpha)^2 t} \\ \Pi_{R2}(p_{r1}^*, p_{n2}^*, w_{rn}^*) = \frac{[(1-\alpha)-\theta(3-\alpha)+\alpha s_r+c(2-\alpha)]^2}{16(2-\alpha)^2 t} \text{ and} \\ \Pi_M(p_{r1}^*, p_{n2}^*, w_{rn}^*) = \frac{[(1-\alpha)(1+\theta)+\alpha s_r-c(2-\alpha)]^2}{8(1-\alpha)(2-\alpha)t}. \end{cases} \tag{28}$$

5.2. Fully-covered market

When $t \le t_2$, the market is fully covered. With demand in (22), we have the results as follows.

Proposition 4. *If retailer 1 offers a full-refund policy and retailer 2 offers a no-refund policy, and if the transportation cost t is sufficiently low ($t \le t_2$),*

(i) *if the two retailers compete in Nash game, the manufacturer sets w_{rn}^{N*} at*

$$w_{rn}^{N*} = \frac{(1-3t+\theta)(1-\alpha)+\alpha s_r}{2-\alpha}; \tag{29}$$

the optimal pricing decisions for the two retailers (p_{r1}^{N*}, p_{n2}^{N*}) become

$$\begin{cases} p_{r1}^{N*} = \frac{(5-2\alpha-3t+\theta)-\alpha s_r}{3(2-\alpha)} \text{ and} \\ p_{n2}^{N*} = \frac{(1-3t+3\theta)(1-\alpha)+2\theta+\alpha s_r}{3(2-\alpha)}; \end{cases} \tag{30}$$

(ii) *if the two retailers compete in Stackelbeg game, the manufacturer sets w_{rn}^{S*} at*

$$w_{rn}^{S*} = \frac{(3-15t+5\theta)(1-\alpha)+3\alpha s_r}{8-5\alpha}; \tag{31}$$

the optimal pricing decisions for two retailers (p_{r1}^{S*}, p_{n2}^{S*}) become

$$\begin{cases} p_{r1}^{S*} = \frac{(7-4\alpha-3t+\theta)-\alpha s_r}{8-5\alpha} \text{ and} \\ p_{n2}^{S*} = \frac{(1-5t+5\theta)(1-\alpha)+2\theta+\alpha s_r}{8-5\alpha}. \end{cases} \tag{32}$$

With Proposition 4, demands are $D_{r1}^{N*} = [(1-\alpha+3t-\theta) + \alpha s_r]/[3t(2-\alpha)]$ and $D_{n2}^{N*} = [(3t-1)(1-\alpha) + \theta - \alpha s_r]/[3t(2-\alpha)]$ in Nash game, which gives when $\underline{\theta}^N < \theta < \overline{\theta}^N$, then $D_{r1}^{N*} > 0$ and $D_{n2}^{N*} > 0$; when $0 < \theta < \underline{\theta}^N$, then $D_{n2}^{N*} = 0$ and $D_{r1}^{N*} = 0$ when $1 > \theta > \overline{\theta}^N$; where $\underline{\theta}^N = (1-\alpha)(1-3t) + \alpha s_r$ and $\overline{\theta}^N = 1-\alpha+3t+\alpha s_r$. In Stackelberg game, demand is $D_{r1}^{S*} = [(1-\alpha+3t-\theta) + \alpha s_r]/[t(8-5\alpha)]$ and $D_{n2}^{S*} = [(5t-1)(1-\alpha) + \theta - \alpha s_r]/[t(8-5\alpha)]$, which gives when $\underline{\theta}^S < \theta < \overline{\theta}^S$, then $D_{r1}^{S*} > 0$ and $D_{n2}^{S*} > 0$; when $0 < \theta < \underline{\theta}^S$, then $D_{n2}^{S*} = 0$ and $D_{r1}^{S*} = 0$ when $1 > \theta > \overline{\theta}^S$, where $\underline{\theta}^S = (1-\alpha)(1-5t) + \alpha s_r$ and $\overline{\theta}^S = 1-\alpha+3t+\alpha s_r$. This suggests that in both games, when the degree to which the customer accepts the no-refund product (θ) is sufficiently low ($0 < \theta < \underline{\theta}^N$ in Nash game or $0 < \theta < \underline{\theta}^S$ in Stackelberg game), no demand will be realized at retailer 2 if it offers a no-refund policy; when θ is moderately high ($\underline{\theta}^N < \theta < \overline{\theta}^N$ in Nash game or $\underline{\theta}^S < \theta < \overline{\theta}^S$ in Stackelberg game), retailers 1 and 2 compete in the market and provide product to customers who prefer a full-refund policy and who prefer a no-refund policy, respectively; retailer 1's demand decreases while retailer 2's demand increases with θ; when θ is sufficiently high ($1 > \theta > \overline{\theta}^N$ in Nash game or $1 > \theta > \overline{\theta}^S$ in Stackelberg game), no demand will be realized at retailer 1 if retailer 2 offers a no-refund policy.

The profits for retailers and the manufacturer are

If two retailers compete in Nash game,

$$\begin{cases} \Pi_{R1}^{N*} = \frac{2(1-\alpha)[(1-\alpha-\theta+3t)+\alpha s_r]^2}{9t(2-\alpha)^2}, \\ \Pi_{R2}^{N*} = \frac{2[(1-\alpha)(1-3t)-\theta+\alpha s_r]^2}{9t(2-\alpha)^2}, \text{ and} \\ \Pi_M^{N*} = \frac{(1-3t+\theta)(1-\alpha)+\alpha s_r-c(2-\alpha)}{2-\alpha}. \end{cases} \tag{33}$$

If two retailers compete in Stackelberg game,

$$\begin{cases} \Pi_{R1}^{S*} = \frac{4(1-\alpha)[(1-\alpha-\theta+3t)+\alpha s_r]^2}{t(8-5\alpha)^2}, \\ \Pi_{R2}^{S*} = \frac{2[(1-\alpha)(1-5t)-\theta+\alpha s_r]^2}{t(8-5\alpha)^2}, \text{ and} \\ \Pi_M^{S*} = \frac{(3-15t+5\theta)(1-\alpha)+3\alpha s_r-c(8-5\alpha)}{8-5\alpha}. \end{cases} \tag{34}$$

Comparing the two games between the two retailers, we can conclude

Corollary 4. *When $0 \le t \le t_2$, $D_{r1}^{S*} < D_{r1}^{N*}$ and $\Pi_{R1}^{S*} > \Pi_{R1}^{N*}$; $D_{n2}^{S*} > D_{n2}^{N*}$ and $\Pi_{R2}^{S*} < \Pi_{R2}^{N*}$ if $\theta < \tilde{\theta}_{n2}$ while $\Pi_{R2}^{S*} > \Pi_{R2}^{N*}$ if $\theta > \tilde{\theta}_{n2}$; $w_{rn}^{S*} < w_{rn}^{N*}$ and $\Pi_M^{S*} < \Pi_M^{N*}$, where $\tilde{\theta}_{n2} = \frac{(1-\alpha)(7-4\alpha)-3t(9-5\alpha)(1-\alpha)+\alpha(7-4\alpha)s_r}{7-4\alpha}$.*

Corollary 4 implies that if retailer 2 as a follower uses a no-refund policy to compete with retailer 1, it will be better off only

when θ is sufficiently high ($\theta > \tilde{\theta}_{n2}$). If $\theta < \tilde{\theta}_{n2}$, retailer 2 is better off if it competes with retailer 1 in Nash game, which also benefits the manufacturer. Since $\Pi_{R2}^{N*} + \Pi_M^{N*} > \Pi_{R2}^{S*} + \Pi_M^{S*}$, the result in corollary 4 suggests that when retailer 2 is competitive in marketing power as compared to retailer 1, if the manufacturer is aware that $\theta > \tilde{\theta}_{n2}$, it can share a portion of its profit so that $\beta(\Pi_M^{N*} - \Pi_M^{S*}) > \Pi_{R2}^{S*} - \Pi_{R2}^{N*}$ to incentivize retailer 2 to compete with retailer 1 in Nash game, rather than as a retailer 1's follower.

6. The impact of θ on retailer 2's returns policy

We now compare the case that both retailers offer a full-refund customer returns policy to the case that retailer 1 offers a full-refund policy while retailer 2 offers a no-refund policy. For both cases, if $t > \max(t_1, t_2)$, the market is partially-covered while is full-covered if $t < \min(t_1, t_2)$, where t_1 and t_2 are given in (8) and (26), respectively. Our discussion in Section 6.1 is for $t > \max(t_1, t_2)$ and in Section 6.2 is for $t < \min(t_1, t_2)$.

6.1. Partially-covered market

Comparing (28) to (7), we have

Corollary 5. *As compared to the case in which both retailers offer a full-refund customer returns policy, when $\theta < \theta_{R1}$, both retailers and the manufacturer will set lower prices (p_{r1}^*, $p_{n2}^*$and w_{rn}^*), and $\Pi_{R1}(p_{r1}^*, p_{n2}^*, w_{rn}^*) > \Pi_{R1}(p_{r1}^*, p_{n2}^*, w_{rn}^*)$, where*

$$\theta_{R1} = 1 - \alpha + \alpha s_r \tag{35}$$

Corollary 5 implies that only when the degree to which the customer accepts the no-refund product (θ) is sufficiently low ($\theta < \theta_{R1}$), retailer 1 can benefit from a no-refund policy that is implemented by retailer 2.

Comparing retailer 2's profit in (28) to (7), we have

Proposition 5. *Only when customer's perceived value of the non-refund product (θ) is higher than a threshold value θ_{R2}, $\Pi_{R2}(p_{r1}^*, p_{n2}^*, w_{rn}^*) > \Pi_{R2}(p_{r1}^*, p_{n2}^*, w_{rn}^*)$, where*

$$\theta_{R2} = \frac{(1-\alpha)[1-\alpha+\alpha s_r + c(2-\alpha)] + \sqrt{1-\alpha}(2-\alpha)(1-\alpha+\alpha s_r - c)]}{(1-\alpha)(3-\alpha)} \tag{36}$$

Also, when $\theta > \theta_{R2}$, $\Pi_{R1}(p_{r1}^*, p_{n2}^*, w_{rn}^*) < \Pi_{R1}(p_{r1}^*, p_{n2}^*, w_{rn}^*)$. Further, $\theta_{R2} > \theta_{R1}$.

Proposition 5 suggests that if retailer 2 is aware of product returns rate (α) and the net salvage value (s_r) at retailer 1, production cost (c) at the manufacturer, and the degree to which the customer accepts the non-refund product in the market (θ), when $\theta > \theta_{R2}$, it is a good option for retailer 2 to implement a no-refund policy to compete with retailer 1; otherwise, implementing a full-refund policy to compete with retailer 1 is a good option. The implication of the above analysis is that when the retailers sell a product with which the customer is very familiar, such as printer cartridges, θ is usually high. If $t > \max(t_1, t_2)$, offering a no-refund policy might be a good option for retailer 2.

For the manufacturer's profit, we have

Corollary 6. *when $\theta > \theta_M$, $\Pi_M(p_{r1}^*, p_{n2}^*, w_{rn}^*) > \Pi_M(p_{r1}^*, p_{n2}^*, w_{rn}^*)$, where $\theta_M > \theta_{R2}$ and*

$$\theta_M = \frac{(1-\alpha+\alpha s_r - c)(\sqrt{4-2\alpha}-1) + c(1-\alpha)}{1-\alpha} \tag{37}$$

Corollary 6 and Proposition 5 suggest that when $\theta > \theta_M$, both retailer 2 and the manufacturer can be better off, but retailer 1 loses profit if retailer 2 offers a no-refund policy, as compared to the case if retailer 2 offers a full-refund policy. Corollary 5 and Proposition 5 show that there exists a range in the degree to which the customer accepts the non-refund product, $\theta_{R1} < \theta < \theta_{R2}$, such that both retailers gain less profit if retailer 2 offers a no-refund policy. With Corollary 6, we see in this range ($\theta_{R1} < \theta < \theta_{R2}$), both retailers, as well as the manufacturer, are better off if retailer 2 offers a full-refund policy to compete with retailer 1.

Corollaries 5 and 6, and Proposition 5 give boundary conditions to define when the new entrant should use a full-refund policy or a no-refund policy to compete with the established retailer and which conditions will benefit the new entrant, the established retailer, or the manufacturer.

6.2. Fully-covered market

For $t < \min(t_1, t_2)$, we denote *RR* and *RN* with brackets as cases in which both retailers offer a full-refund policy and in which retailer 1 offers a full-refund policy while retailer 2 offers a no-refund policy, respectively. To identify boundary conditions for when the new entrant should use a full-refund policy or a no-refund policy to compete with the established retailer, we first define several boundary values as follows.

(i) Letting θ_1^N be the boundary value of $\Pi_{R2}^N(RR) = \Pi_{R2}^N(RN)$, (12) and (33) give

$$\theta_1^N = [2(1-3t)(1-\alpha) + 2\alpha s_r + 3t(2-\alpha)\sqrt{1-\alpha}]/2 \tag{38}$$

(ii) Letting θ_1^S be the boundary value of $\Pi_{R2}^S(RR) = \Pi_{R2}^S(RN)$, (14) and (34) give

$$\theta_1^S = [8(1-5t)(1-\alpha) + 8\alpha s_r + 5t(8-5\alpha)\sqrt{1-\alpha}]/8 \tag{39}$$

(iii) Letting θ_2^N be the boundary value of $\Pi_{R1}^N(RR) = \Pi_{R1}^N(RN)$, (12) and (33) give

$$\theta_2^N = [2(1-\alpha) + 2\alpha s_r + 3t\alpha]/2 \tag{40}$$

(iv) Letting θ_2^S be the boundary value of $\Pi_{R1}^S(RR) = \Pi_{R1}^S(RN)$, (14) and (34) give

$$\theta_2^S = [8(1-\alpha) + 15t\alpha + 8\alpha s_r]/8 \tag{41}$$

(v) Letting θ_3^N be the boundary value of $\Pi_M^N(RR) = \Pi_M^N(RN)$, (19) and (33) give

$$\theta_3^N = [2(1-\alpha) + 2\alpha s_r + 3t\alpha]/2 \tag{42}$$

(vi) Letting θ_3^S be the boundary value of $\Pi_M^S(RR) = \Pi_M^S(RN)$, (20) and (34) give

$$\theta_3^S = [8(1-\alpha) + 15t\alpha + 8\alpha s_r]/8 \tag{43}$$

Notice that $\theta_2^N = \theta_3^N$ and $\theta_2^S = \theta_3^S$. We can conclude

Corollary 7.

(i) *if $\theta < \theta_2^N$, then $\Pi_{R1}^N(RN) > \Pi_{R1}^N(RR)$ while $\Pi_{R1}^S(RN) > \Pi_{R1}^S(RR)$ if $\theta < \theta_2^S$;*
(ii) *if $\theta > \theta_1^N$, then $\Pi_{R2}^N(RN) > \Pi_{R2}^N(RR)$ while $\Pi_{R2}^S(RN) > \Pi_{R2}^S(RR)$ if $\theta > \theta_2^S$;*
(iii) *if $\theta > \theta_2^N = \theta_3^N$, then $\Pi_M^N(RN) > \Pi_M^N(RR)$ while $\Pi_M^S(RN) > \Pi_M^S(RR)$ if $\theta > \theta_2^S = \theta_3^S$.*

(38)–(41) give $\theta_1^N - \theta_2^N = -[3t(2-\alpha)(1-\sqrt{1-\alpha})]/2 < 0$, and $\theta_1^S - \theta_2^S = -[5t(8-5\alpha)(1-\sqrt{1-\alpha})]/8 < 0$. Therefore, we have $\theta_1^N < \theta_2^N = \theta_3^N$, $\theta_1^S < \theta_2^S = \theta_3^S$. With Corollary 7, we can have the following results.

Proposition 6.

(i) *if* $0<\theta<\theta_1^N$, *then* $\Pi_{R1}^N(RN)>\Pi_{R1}^N(RR)$ *while* $\Pi_{R1}^S(RN)>\Pi_{R1}^S(RR)$ *if* $0<\theta<\theta_1^S$;
(ii) *if* $\theta_1^N<\theta<\theta_2^N$, *then* $\Pi_{R2}^N(RN)>\Pi_{R2}^N(RR)$ *and* $\Pi_{R1}^N(RN)>\Pi_{R1}^N(RR)$; *if* $\theta_1^S<\theta<\theta_2^S$, *then* $\Pi_{R2}^S(RN)>\Pi_{R2}^S(RR)$ *and* $\Pi_{R1}^S(RN)>\Pi_{R1}^S(RR)$;
(iii) *if* $\theta_2^N<\theta<1$, *then* $\Pi_{R2}^N(RN)>\Pi_{R2}^N(RR)$ *and* $\Pi_M^N(RN)>\Pi_M^N(RR)$; *if* $\theta_2^S<\theta<1$, *then* $\Pi_{R2}^S(RN)>\Pi_{R2}^S(RR)$ *and* $\Pi_M^S(RN)>\Pi_M^S(RR)$.

Proposition 6 suggests that, as compared to the case in which both retailers offer a full-refund customer returns policy, only retailer 1 gains more profit if retailer 2 competes with retailer 1 by offering a no-refund policy when $0<\theta<\theta_1^N$ for Nash game and when $0<\theta<\theta_1^S$ for Stackelberg game. This implies that when the market is fully covered (i.e., t is sufficiently low), if the customer's view of the value of the product sold with a no-refund policy (θ) is low, more customers will buy the product from retailer 1 (i.e., demand decreases in θ for retailer 1). Therefore, retailer 1 benefits from retailer 2's no-refund policy. As the customer's value of the no-refund product increases, both retailers profit, while the manufacturer gains less profit if $\theta_1^N<\theta<\theta_2^N$ for Nash game and if $\theta_1^S<\theta<\theta_2^S$ for Stackelberg game between two retailers. Retailer 2 enhances profit through savings from not handling returns or paying refunds to customers, although it sets a lower retail price, which benefits customers who purchase the product. Retailer 2 obtains extra profit through setting a lower retail price to attract more customers. Profits increase for retailer 2 and the manufacturer but not for retailer 1 if retailer 2 offers a no-refund policy when $\theta_2^N<\theta<1$ for Nash game and when $\theta_2^S<\theta<1$ for Stackelberg game. This implies that when customers' view on the value of product sold with a no-refund policy (θ) is higher than θ_2^N for Nash game or higher than θ_2^S for Stackelberg game, the manufacturer will set a higher wholesale price, which enhances its profit. Both retailers raise selling price; retailer 2's demand increases, which increases its profit.

7. Numerical examples

We use numerical examples to illustrate the impact of customer returns on retailer 2's returns policy, and the profits of the two retailers and the manufacturer. We set $\theta=0.8$, $s_r=0.05$, $c=0.15$. Fig. 2 shows the change in t_1 and t_2 with α. Since $t_1\big|_{\alpha=0}>t_2\big|_{\alpha=0}$ for $\theta<1$ and both t_1 and t_2 decrease with α [from (8) and (26)], $t_1>t_2$ until $\alpha=0.667$ in this example.

Fig. 3 shows the change in θ_{R1} [in (35)], θ_{R2} [in (36)], and θ_M [in (37)] with α for $t=0.4>\max(t_1,t_2)$. As compared to the case when both retailers offer a full-refund policy, for $t=0.4>\max(t_1,t_2)$, and $\theta=0.8$ in this example, when customer returns rate is small ($\alpha<0.21$), retailer 1 is better off if retailer 2 offers a no-refund policy; when $0.21<\alpha<0.28$, profit increases for both retailers and the manufacturer if retailer 2 implements a full-refund policy; when $0.28<\alpha<0.35$, only retailer 2 benefits if it offers a no-refund policy; when $\alpha>0.35$, both retailer 2 and the manufacturer gain more profit, but retailer 1 gains less profit if retailer 2 offers a no-refund policy. The example illustrates that when retailer 2 is aware of θ, based on information on customer returns rate (α), it should carefully check if $\theta>\theta_{R2}$ (Proposition 5) before deciding which customer returns policy, a full-refund or a no-refund policy, should be offered as a strategy to compete with retailer 1.

For $t=0.2\le\min(t_1,t_2)$, Fig. 4(a) shows the change in θ_1^N, θ_2^N, and θ_3^N, and Fig. 4(b) shows the change of θ_1^S, θ_2^S, and θ_3^S with customer returns rate (α). For $\alpha=0.8$, $t_1=0.225$ and $t_2=0.275$. Fig. 5(a) shows the change in θ_1^N, θ_2^N, and θ_3^N, and Fig. 5(b) shows the change in θ_1^S, θ_2^S, and θ_3^S with transportation cost (t), where $t\le\min(t_1,t_2)=0.225$.

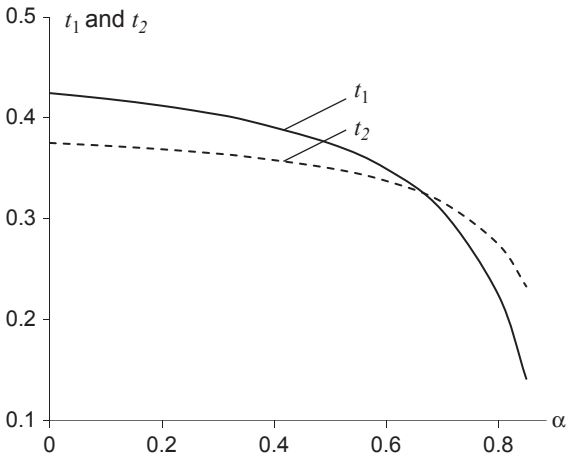


Fig. 2. The changes of t_1 and t_2 with α.

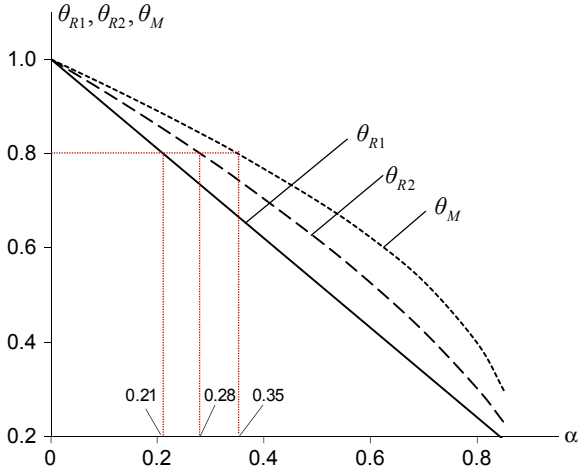


Fig. 3. Changes of θ_{R1}, θ_{R2}, and θ_M with α for $t=0.4$.

As compared to the case in which both retailers offer a full-refund customer returns policy, Figs. 4 and 5 show that if retailer 2 knows that θ falls in the shadowed region I, both retailer 2 and the manufacturer can increase their profits, but retailer 1 gains less profit; if retailer 2 knows that θ falls in the shadowed region II, both retailers can increase profits, but the manufacturer gains less profit; if the retailer 2 is aware that θ falls in the shadowed region III, only retailer 1 can increase profit. Fig. 4 shows that as customer returns rate (α) increases, both regions I and II expand, but region III shrinks significantly. This implies that as α increases, there is more space in which both retailer 2 and the manufacturer increase profits; there is also more space in which both retailers increase profits, but less space in which only retailer 1 gains more profit. Fig. 5 shows that as t increases, region I shrinks while region II expands, suggesting that there is more space in t for both retailers to gain more profits, and less space in t for both retailer 2 and the manufacturer to gain more profits.

In this paper, we assume that two retailers have identical returns rates (α) and net salvage values (s_r). We now use a numerical example to illustrate the impact of different net salvage values of retailer 1 (s_{r1}) and of retailer 2 (s_{r2}) as well as different customer returns rates (α_1 and α_2 for retailers 1 and 2, respectively) on their pricing decisions and profits. We set $s_{r1}=0.05$, $c=0.15$, $t=0.3$ and $\alpha_1=0.25$. Figs. 6–8 show the change in prices and profits for the manufacturer and two retailers with s_{r2} for different customer returns rate at retailer 2 (α_2) when both retailers offer full-refund policies, for both Stackelberg game and Nash game.

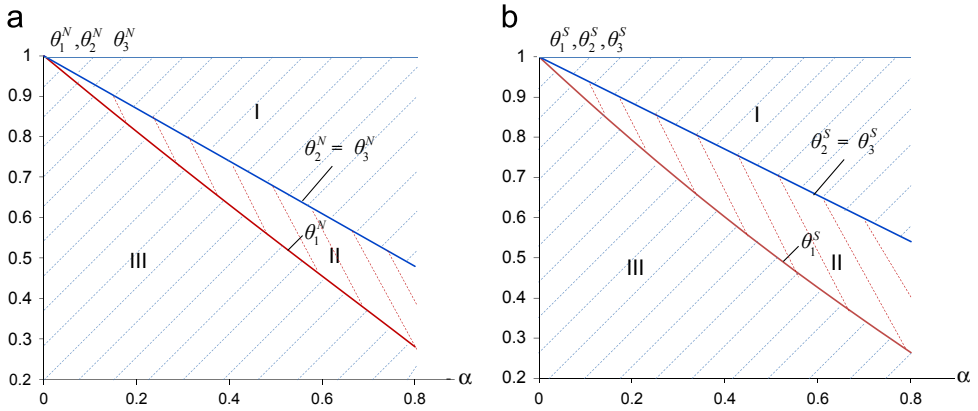


Fig. 4. Change in θ_1^N, θ_2^N, θ_3^N, θ_1^S, θ_2^S, and θ_3^S with α. (a) Change in θ_1^N, θ_2^N and θ_3^N with α and (b) Change in θ_1^S, θ_2^S, and θ_3^S with α.

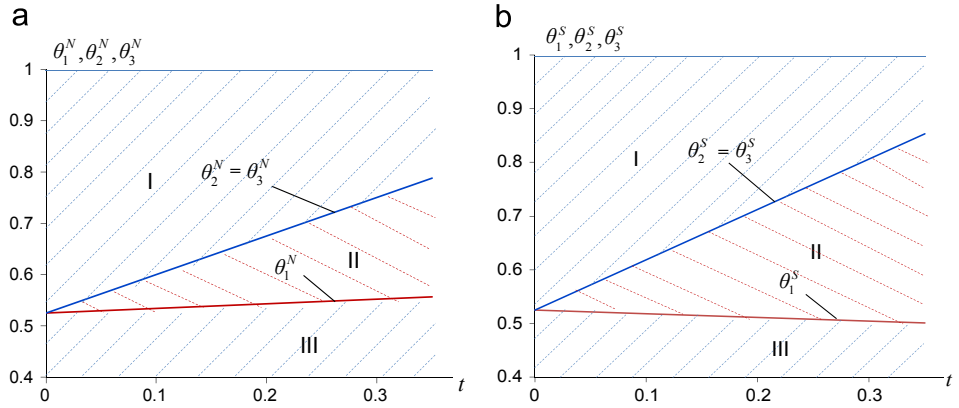


Fig. 5. Change in θ_1^N, θ_2^N, θ_3^N, θ_1^S, θ_2^S, and θ_3^S with t. (a) Change in θ_1^N, θ_2^N, and θ_3^N with t. (b) Change in θ_1^S, θ_2^S, and θ_3^S with t.

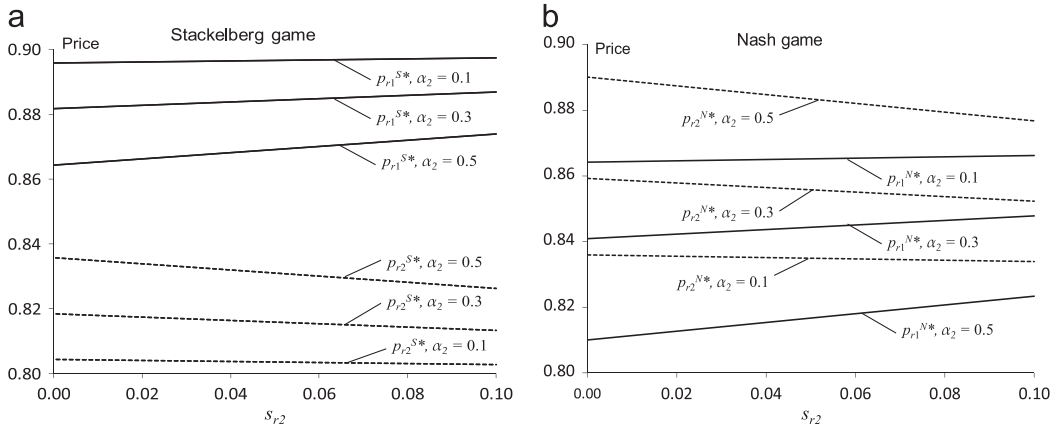


Fig. 6. The changes in prices with retailer 2's net salvage value (s_{r2}) for different customer returns rate (α_2) when both retailers offer full-refund policies. (a) Stackelberg game and (b) Nash game.

Figs. 6 and 7 show that for both Stackelberg and Nash games, as s_{r2} increases (i.e., as the retailer 2 is getting more efficient in handling customer returns), retailer 2 will set a slightly lower price and gain slightly high profit. To respond the increasing in s_{r2}, retailer 1 will slightly raise price and incur a loss in profit. In addition, as the increase of customer returns rate at retailer 2 (α_2), retailer 2 will raise price and reduce the order quantity to mitigate the negative impact of customer returns. In the same time, retail 1 will cut price to attract more customers and thus benefit from the increased customer returns at retailer 2. Both retailers prefer Stackelberg game to Nash game. Furthermore, this numerical study (Fig. 8) also shows that as s_{r2} increases, the manufacturer will raise the wholesale price, resulting in more profit. The manufacturer incurs a profit loss when retailer 2 faces more customer returns. The manufacturer can set a higher wholesale price when two retailers compete in Nash game than when they compete in Stackelberg game which leads to a more profit.

8. Conclusion

In this paper, we discuss how a new entrant retailer can use customer returns policies, specifically a full-refund policy or a no-refund policy, to compete with an established retailer who offers a full-refund customer returns policy.

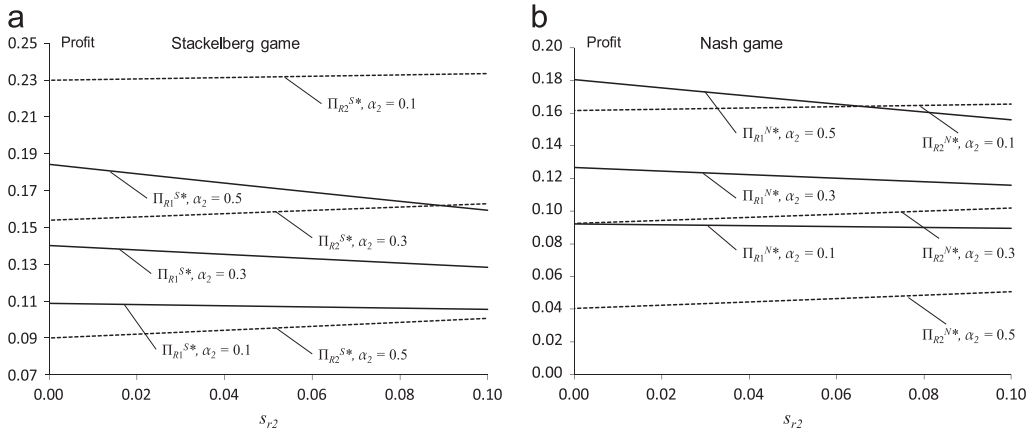


Fig. 7. The changes in profits with retailer 2's net salvage value (s_{r2}) for different customer returns rate (α_2) when both retailers offer full-refund policies. (a) Stackelberg game and (b) Nash game.

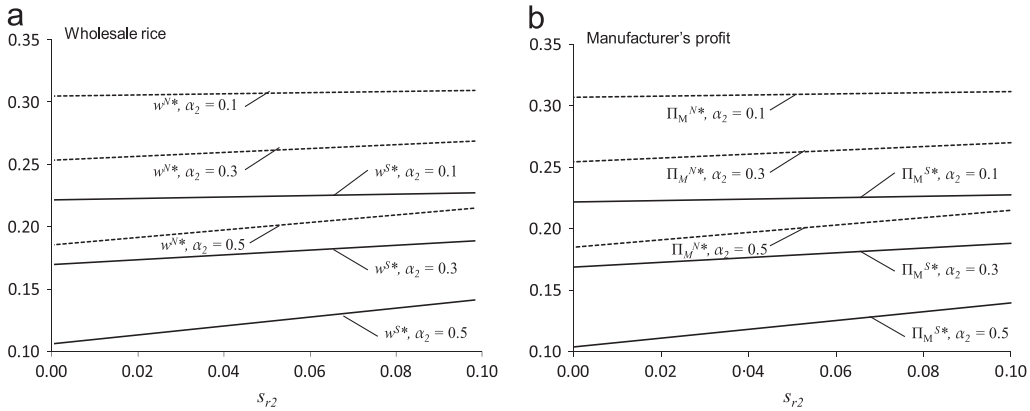


Fig. 8. The changes in wholesale prices and manufacturer's profits with retailer 2's net salvage value (s_{r2}) for different α_2 when both retailers offer full-refund policies. (a) wholesale prices and (b) profits.

Providing a customer returns policy is usually regarded as a service to customers. This research, however, provides new insights for competing retailers facing customer returns in a supply chain. We show that implementing different customer returns policies can be used as a competitive strategy in the market to enhance a retailer's profit, while meeting the needs of distinct customer groups. Specifically, we indentify the boundary conditions for when the new entrant retailer should use a full-refund policy or a no-refund policy to compete with an established retailer.

We show that, using a no-refund policy is a good option for the new entrant retailer when the transportation cost is sufficiently high (i.e., the market is partially-covered), only when the degree to which the customer accepts the no-refund product is higher than a threshold value. In addition, there exists a range of customer acceptance of the no-refund product ($\theta_{R1} < \theta < \theta_{R2}$) such that both retailers, as well as the manufacturer, are better off if new entrant retailer offers a full-refund policy to compete with the established retailer.

We also show when the transportation cost is sufficiently low (i.e., the market is fully-covered), if the new entrant retailer decides to use a full-refund policy to compete with the established retailer, both retailers' pricing decisions are independent of customer returns rate and net salvage value.

The present research also has implications for pricing based on type of product. For example, our results suggest that when a new or unfamiliar product is marketed in the face of uncertain customer expectations (the lower θ case), the new entrant retailer should use a full-refund policy and set a higher retail price. If the product is mature and the uncertainty of meeting the customer's expectations is lower (the higher θ case), selling the product using a no-refund policy is a competitive strategy for the new entrant retailer, allowing competition with the established retailer. In this way, the new entrant retailer can attract customers who are familiar with the product but prefer a lower price.

The research shows that it is critical for the new entrant retailer competing with an established retailer to understand the degree to which the customer accepts the no-refund product (θ) before deciding on a customer returns policy. Comprehensive marketing research is needed to understand the customer's preference (a low price with a no-refund policy vs. a high price with a full-refund policy) so that the new entrant retailer can obtain a relatively precise measure of the degree to which the customer accepts the no-refund product.

This research also shows that competing using customer returns policies can be a profit enhancement strategy and highlights the need for the supply chain that is facing customer returns to take customer returns into consideration in pricing decisions.

In this paper, we raise some notes of caution for the manufacturer; the decision of the new entrant retailer on customer returns policies and pricing games may not benefit the manufacturer. The manufacturer might need a side contract instead of the wholesale price only contract to incentivize the new entrant retailer, so that both can benefit from the decision of the new entrant retailer. This situation should be addressed in future research.

Acknowledgments

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Appendix

Proof of Proposition 1:

Proof. With demands in (1), for a given w, taking partial derivatives of (3) and (4) w.r.t. $p_{r1}$and p_{r2}, respectively:

$\partial\Pi_{R1}/\partial p_{r1}=[1-\alpha-2p_{r1}(1-\alpha)-\alpha s_r+w]/t$ and

$\partial\Pi_{R2}/\partial p_{r2}=[1-\alpha-2p_{r2}(1-\alpha)-\alpha s_r+w]/t$

$\partial\Pi_{R1}{}^2/\partial^2 p_{r1}=-2(1-\alpha)/t<0$ and

$\partial\Pi_{R2}{}^2/\partial^2 p_{r2}=-2(1-\alpha)/t<0$

implying that there exist unique optimal solutions to p_{r1} and p_{r2} which are given by setting $\partial\Pi_{R1}/\partial p_{r1}$ and $\partial\Pi_{R2}/\partial p_{r2}$ to be equal to 0:

$$p_{r1}=p_{r2}=(1-\alpha-\alpha s_r+w)/[2(1-\alpha)] \tag{A1}$$

with (1) and substituting (A1) into (5)

$$\Pi_M=[(w-c)(1-\alpha+\alpha s_r-w)]/t \tag{A2}$$

Taking partial derivatives of (A2) w.r.t. w:

$\partial\Pi_M/\partial w=[(1-\alpha)(1-\alpha+\alpha s_r-2w+c)]/t$ and

$\partial\Pi_M{}^2/\partial^2 w=-2(1-\alpha)/t<0$

$\partial\Pi_M{}^2/\partial^2 w<0$

suggesting that there exists a unique optimal w_{rr}^* by setting $\partial\Pi_M/\partial w$ to be 0, which gives

$$w_{rr}^*=[1-\alpha+\alpha s_r+c]/2 \tag{A3}$$

With (A3), p_{r1}^* and p_{r2}^* in (A1) are $p_{r1}^*=p_{r2}^*=[3(1-\alpha)-\alpha s_r+c]/[4(1-\alpha)]$.

Proof of Corollary 1

Proof. In Nash equilibrium, for a given w, with demand in (2), taking partial derivatives of profit functions of two retailers in (3) and (4) w.r.t. p_{r1} and p_{r2}, respectively:

$\partial\Pi_{R1}/\partial p_{r1}=[-(2p_{r1}-p_{r2}-t)(1-\alpha)-\alpha s_r+w]/(2t)$ and

$\frac{\partial\Pi_{R2}}{\partial p_{r2}}=\frac{-(2p_{r2}-p_{r1}-t)(1-\alpha)-\alpha s_r+w}{2t}$

$\frac{\partial\Pi_{R1}{}^2}{\partial^2 p_{r1}}=\frac{\partial\Pi_{R2}{}^2}{\partial^2 p_{r2}}=-(1-\alpha)/t<0$ and

$\frac{\partial\Pi_{R1}{}^2}{\partial p_{r1}\partial p_{r2}}=\frac{\partial\Pi_{R2}{}^2}{\partial p_{r2}\partial p_{r1}}=(1-\alpha)/(2t)$ give

$\frac{\partial\Pi_{R1}{}^2}{\partial^2 p_{r1}}\frac{\partial\Pi_{R2}{}^2}{\partial^2 p_{r2}}-\frac{\partial\Pi_{R2}{}^2}{\partial p_{r2}\partial p_{r1}}\frac{\partial\Pi_{R1}{}^2}{\partial p_{r1}\partial p_{r2}}=\frac{3}{4(1-\alpha)^2t^2}>0$

implying that there exists a unique optimal pair (p_{r1},p_{r2}) which is given by setting $\partial\Pi_{R1}/\partial p_{r1}$ and $\partial\Pi_{R2}/\partial p_{r2}$ to be equal to 0 simultaneously:

$$p_{r1}^{N*}=p_{r2}^{N*}=[t(1-\alpha)-\alpha s_r+w]/(1-\alpha) \tag{A4}$$

In Stackelberg equilibrium, for a given w, with demand in (2), taking partial derivatives of profit function of retailer 2 w.r.t. p_{r2}:

$\partial\Pi_{R2}/\partial p_{r2}=[-(2p_{r2}-p_{r1}-t)(1-\alpha)-\alpha s_r+w]/2t$ and

$\partial\Pi_{R1}{}^2/\partial^2 p_{r1}<0$

suggesting that for a given p_{r1}, there exists a unique p_{r2}^{S*} which is given by setting $\partial\Pi_{R2}/\partial p_{r2}=0$:

$$p_{r2}^{S*}=[(p_{r1}+t)(1-\alpha)-\alpha s_r+w]/[2(1-\alpha)] \tag{A5}$$

with (A5), demand in (2), taking partial derivatives of (3) w.r.t. p_{r1}:

$\frac{\partial\Pi_{R1}}{\partial p_{r1}}=\frac{-(2p_{r1}+3t)(1-\alpha)-2\alpha s_r+2w}{4t}$ and

$\frac{\partial\Pi_{R1}{}^2}{\partial^2 p_{r1}}=-\frac{1-\alpha}{2t}<0$

implying that there exists a unique optimal p_{r2}^{S*} which is given by setting $\partial\Pi_{R1}/\partial p_{r1}=0$:

$$p_{r1}^{S*}=[3t(1-\alpha)-2s_r\alpha+2w]/[2(1-\alpha)] \tag{A6}$$

With p_{r2}^{S*} in (A6), (A5) becomes

$$p_{r2}^{S*}=[5t(1-\alpha)-4s_r\alpha+4w]/[4(1-\alpha)] \tag{A7}$$

Proof of Proposition 2:

Proof. In Nash equilibrium, with respective demands given in (11), from the manufacturer's profit function given in (5), we see that demands in (11) are independent of w. From $t\le t_1=2-p_{r1}-p_{r2}$, we have $w\le[(2-3t)(1-\alpha)+2\alpha s_r]/2$. Since manufacturer's profit increases with w, thus we have

$$w_{rr}^{N*}=[(2-3t)(1-\alpha)+2\alpha s_r]/2 \tag{A8}$$

Substituting (A8) into (A4), we have $p_{r1}^{N*}=p_{r2}^{N*}=(2-t)/2$.

In Stackelberg equilibrium, with respective demands given in (13), from the manufacturer's profit function given in (5), we see that demands in (11) are independent of w. From $t\le t_1=2-p_{r1}-p_{r2}$, we have $w\le[(8-15t)(1-\alpha)+8\alpha s_r]/8$. Since manufacturer's profit increases with w, thus we have

$$w_{rr}^{S*}=[(8-15t)(1-\alpha)+8\alpha s_r]/8 \tag{A9}$$

Substituting (A9) into (A6) and (A7), we have $p_{r1}^{S*}=(8-3t)/8$ and $p_{r2}^{S*}=(8-5t)/8$.

Proof of Proposition 3:

Proof. With demands in (21), for a given w, taking partial derivatives of Π_{R1} and Π_{R2} in (23) w.r.t. p_{r1} and p_{n2}, respectively.

$\partial\Pi_{R1}/\partial p_{r1}=[1-\alpha-2p_{r1}(1-\alpha)-\alpha s_r+w]/t$ and

$\partial\Pi_{R2}/\partial p_{r2}=[-2p_{n2}+\theta+w]/t$

$\partial\Pi_{R1}{}^2/\partial^2 p_{r1}=-2(1-\alpha)/t<0$ and $\partial\Pi_{R2}{}^2/\partial^2 p_{r2}=-2/t<0$

implying that there exist unique optimal solutions to p_{r1} and p_{n2}, which are given by setting $\partial\Pi_{R1}/\partial p_{r1}$ and $\partial\Pi_{R2}/\partial p_{r2}$ to be equal to 0:

$$\begin{cases} p_{r1}=\frac{1-\alpha-\alpha s_r+w}{2(1-\alpha)} \text{ and} \\ p_{n2}=\frac{\theta+w}{2}. \end{cases} \tag{A10}$$

With (21) and substituting (A10) into (24):

$$\Pi_M=[(w-c)(1-\alpha+\alpha s_r+(\theta-w)(1-\alpha)-w)]/[2t(1-\alpha)] \tag{A11}$$

Taking partial derivatives of (A11) w.r.t.

$w:\partial\Pi_M/\partial w=[(1-\alpha)(1+\theta)+\alpha s_r-(2w-c)(2-\alpha)]/[2t(1-\alpha)]$ *and*

$\partial\Pi_M{}^2/\partial^2 w=-2(1-\alpha)(1-\alpha)/t<0$ $\partial\Pi_M{}^2/\partial^2 w<0$

suggests that there exists a unique optimal w_{rn}^* by setting $\partial\Pi_M/\partial w=0$, which gives

$$w_{rn}^*=[(1-\alpha)(1+\theta)+\alpha s_r+c(2-\alpha)]/[2(2-\alpha)] \tag{A12}$$

With (A12), we have p_{r1}^* and p_{n2}^* in (A10):

$$\begin{cases} p_{r1}^*=[(1-\alpha)(2+\theta)+[(1-\alpha)-\alpha s_r](3-2\alpha)+c(2-\alpha)]/[4(2-\alpha)(1-\alpha)], \text{and} \\ p_{n2}^*=[(1-\alpha)+\theta(5-3\alpha)+\alpha s_r+c(2-\alpha)]/[4(2-\alpha)]. \end{cases}$$

Proof of Corollary 4:

Proof.

$$D_{r1}^{S*}-D_{r1}^{N*}=-2(1-\alpha)[(1-\alpha-\theta+3t)+\alpha s_r]/[3t(8-5\alpha)(2-\alpha)]<0$$

$$\Pi_{R1}^{S*}-\Pi_{R1}^{N*}=2(1-\alpha)(8+8\alpha-7\alpha^2)[(1-\alpha-\theta+3t)+\alpha s_r]^2/[9t(8-5\alpha)^2(2-\alpha)^2]>0$$

$$D_{n2}^{S*}-D_{n2}^{N*}=2(1-\alpha)[(1-\alpha-\theta+3t)+\alpha s_r]/[3t(8-5\alpha)(2-\alpha)]>0$$

$$\Pi_{R2}^{S*}-\Pi_{R2}^{N*}=\frac{\begin{pmatrix}8(1-\alpha)[(-1+\alpha+\theta-\alpha s_r)(7-4\alpha)+3t(9-5\alpha)(1-\alpha)]\\ \times[(1-\alpha-\theta+3t)+\alpha s_r]\end{pmatrix}}{9t(8-5\alpha)^2(2-\alpha)^2}$$

gives that when

$$\theta<\tilde{\theta}_{n2}=[(1-\alpha)(7-4\alpha)-3t(9-5\alpha)(1-\alpha)+\alpha(7-4\alpha)s_r]/(7-4\alpha)$$

otherwise

$$\Pi_{R2}^{S*}>\Pi_{R2}^{N*}$$

$$w_{rn}^{S*}-w_{rn}^{N*}=-2(1-\alpha)[(1-\alpha-\theta+3t)+\alpha s_r]/[3(8-5\alpha)(2-\alpha)]<0$$

$$\Pi_M^{S*}-\Pi_M^{N*}=-2(1-\alpha)[(1-\alpha-\theta+3t)+\alpha s_r]/[(8-5\alpha)(2-\alpha)]<0$$

Proof of Proposition 4:

Proof. In Nash equilibrium, for a given w, with respective demands given in (22), taking partial derivatives of Π_{R1} and Π_{R2}in (23) w.r.t.p_{r1} and p_{n2}, respectively:

$\partial\Pi_{R1}/\partial p_{r1}=[(1-\theta+t+p_{n2})(1-\alpha)-2p_{r1}(1-\alpha)-\alpha s_r+w]/(2t)$ and

$\partial\Pi_{R2}/\partial p_{n2}=[-2p_{n2}+p_{r1}+\theta+t-1+w]/(2t)$

$\partial{\Pi_{R1}}^2/\partial^2 p_{r1}=-(1-\alpha)/t<0$, $\partial{\Pi_{R2}}^2/\partial^2 p_{n2}=-1/t<0$

$\partial{\Pi_{R1}}^2/(\partial p_{r1}\partial p_{n2})=\partial{\Pi_{R2}}^2/(\partial p_{n2}\partial p_{r1})=(1-\alpha)/(2t)$, and

$$\frac{\partial{\Pi_{R1}}^2}{\partial^2 p_{r1}}\frac{\partial{\Pi_{R2}}^2}{\partial^2 p_{n2}}-\frac{\partial{\Pi_{R1}}^2}{\partial p_{r1}\partial p_{n2}}\frac{\partial{\Pi_{R2}}^2}{\partial p_{n2}\partial p_{r1}}=(3-\alpha)(1-\alpha)/(4t^2)>0$$

implying that there exists a unique optimal solution pair (p_{r1}^*, p_{n2}^*) which are given by solving $\frac{\partial\Pi_{R1}}{\partial p_{r1}}=0$ and $\frac{\partial\Pi_{R2}}{\partial p_{n2}}=0$:

$$\begin{cases}p_{r1}=[(1-\theta+3t)(1-\alpha)-2\alpha s_r+w(3-\alpha)]/[3(1-\alpha)] \text{ and}\\ p_{n2}=[-(1-\theta-3t)(1-\alpha)-\alpha s_r+w(3-2\alpha)]/[3(1-\alpha)].\end{cases}\quad(A13)$$

Since when $t\le t_2$, the market is fully-covered and the total demand is 1. Eq. (24) gives that the manufacturer's profit increases with w. From $t\le t_2=1-p_{r1}+\theta-p_{n2}$, we have $w\le[(1-3t+\theta)(1-\alpha)+\alpha s_r]/(2-\alpha)$. Thus we have

$$w_{rn}^{N*}=[(1-3t+\theta)(1-\alpha)+\alpha s_r]/(2-\alpha)\quad(A14)$$

Substituting (A14) into (A13), we have $p_{r1}^{N*}=[(5-2\alpha-3t+\theta)-\alpha s_r]/[3(2-\alpha)]$ and

$$p_{n2}^{N*}=[(1-3t+3\theta)(1-\alpha)+2\theta+\alpha s_r]/[3(2-\alpha)].$$

In Stackelberg equilibrium, for a given w, with respective demands given in (22), taking partial derivatives of Π_{R2} in (23) w.r.t. p_{n2}:
$\partial\Pi_{R2}/\partial p_{n2}=[-2p_{n2}+p_{r1}+\theta+t-1+w]/(2t)$.
$\partial{\Pi_{R2}}^2/\partial^2 p_{n2}=-1/t<0$, suggesting that for a given p_{r1}, there exists a unique p_{n2} which is given by setting $\partial\Pi_{R2}/\partial p_{n2}=0$:

$$p_{n2}^{S*}=(p_{r1}+w-1+t+\theta)/2\quad(A15)$$

With p_{n2}^{N*} in (A15), for a given w, taking partial derivatives of Π_{R1} in (23) w.r.t. p_{r1}:

$\partial\Pi_{R1}/\partial p_{r1}=[(1-\theta+3t-2p_{r1})(1-\alpha)+w(2-\alpha)-\alpha s_r]/(4t)$ and $\partial\Pi R1^2/\partial^2 p_{r1}=-(1-\alpha)/(2t)<0$, implying that there exists a unique optimal solution p_{r1}^* which is given by setting $\partial\Pi_{R2}/\partial p_{n2}=0$:

$$p_{r1}^{S*}=[(1-\theta+3t)(1-\alpha)+w(2-\alpha)-\alpha s_r]/[2(1-\alpha)]\quad(A16)$$

With (A16), (A15) becomes

$$p_{n2}^{S*}=[(-1+\theta+5t)(1-\alpha)+w(4-3\alpha)-\alpha s_r]/[4(1-\alpha)]\quad(A17)$$

Since when $t\le t_2$, the market is fully-covered and the total demand is 1. Eq. (24) shows that the manufacturer's profit increases with w. From $t\le t_2=1-p_{r1}+\theta-p_{n2}$, we have $w\le[(3-15t+5\theta)(1-\alpha)+3\alpha s_r]/(8-5\alpha)$, leading to

$$w_{rn}^{S*}=[(3-15t+5\theta)(1-\alpha)+3\alpha s_r]/(8-5\alpha)\quad(A18)$$

Substituting (A18) into (A16) and (A17), then $p_{r1}^S*=[(7-4\alpha-3t+\theta)-\alpha s_r]/[8-5\alpha]$ and

$$p_{n2}^{S*}=[(1-5t+5\theta)(1-\alpha)+2\theta+\alpha s_r]/[8-5\alpha].$$

Proof of Corollary 5:

Proof. comparing (7) and (28), we see that when $\theta<\theta_{R1}=1-\alpha+\alpha s_r$, $\Pi_{R1}(p_{r1}^*, p_{n2}^*, w_{rn}^*)>\Pi_{R1}(p_{r1}^*, p_{n2}^*, w_{rn}^*)$.

Proof of Proposition 5:

Proof. comparing (7) to (28), it is easy to see that when $\theta>\theta_{R2}$,$\Pi_{R2}(p_{r1}^*, p_{n2}^*, w_{rn}^*)>\Pi_{R2}(p_{r1}^*, p_{n2}^*, w_{rn}^*)$. In addition, $\theta_{R2}-\theta_{R1}=[(1-\alpha)(\sqrt{1-\alpha}-1+\alpha)(1-\alpha+\alpha s_r-c)]/[(1-\alpha)(3-\alpha)]>0$ if $\alpha<\overline{\alpha}_{rr}=(1-c)/(1-s_r)$, which is a sufficient condition for generating positive demand in Section 4.1.

Proof of Corollary 6:

Proof. Comparing θ_M to $\overline{\theta}$, $\theta_M>\overline{\theta}$ if, which is a sufficient condition for generating positive demand in Section 4.1.

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